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Python Trading Guide: MACD

Implementing MACD in Python



Okan Yenigün

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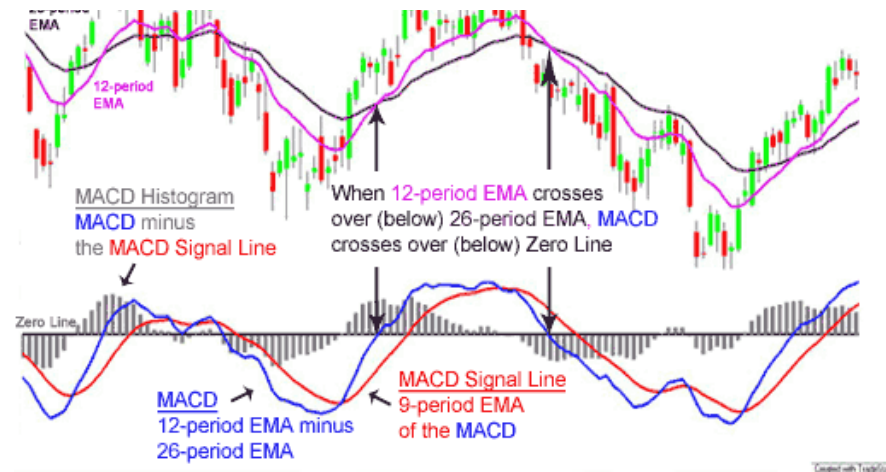


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The Moving Average Convergence Divergence (MACD) is a widely used indicator in algorithmic trading and technical analysis. It consists of three main components:

- **MACD Line** is the difference between two exponential moving averages (EMAs) of a security's price, typically the 12-day and 26-day EMAs.
- **Signal Line** is an EMA of the MACD line, usually with a period of 9 days.
- **Histogram** represents the difference between the MACD line and the Signal line.





MACD. [Source](#)

A common trading signal is when the MACD line crosses above or below the Signal line. A crossover above the Signal line is viewed as bullish, while a crossover below is bearish.



[Source](#)

When the MACD line diverges from the price of the security, it suggests a possible reversal. For instance, if the price is making new highs but the MACD is not, it could indicate weakening momentum.

The MACD line crossing above or below zero is also a signal. Above zero for a sustained period of time suggests an upward trend, while below zero indicates a downward trend.

Like all indicators, MACD is not foolproof. It's a lagging indicator, meaning it's based on past prices and may not always predict future price movements accurately. It can also produce false signals in a sideways or ranging market.

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Python Code

Let's make a demonstration. We can use the `yfinance` library to collect 'TSLA' stock prices. And we can utilize `ta` library to compute MACD.

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pip install yfinance, ta

```
import yfinance as yf
import matplotlib.pyplot as plt
import ta

data = yf.download('TSLA', start='2020-01-01', end='2023-01-01')
data.head()
```

[*****100%*****] 1 of 1 completed

	Open	High	Low	Close	Adj Close	Volume
Date	~/Desktop/codes/python__general/c.py					
2020-01-02	28.299999	28.713333	28.114000	28.684000	28.684000	142981500
2020-01-03	29.366667	30.266666	29.128000	29.534000	29.534000	266677500
2020-01-06	29.364668	30.104000	29.333332	30.102667	30.102667	151995000
2020-01-07	30.760000	31.441999	30.224001	31.270666	31.270666	268231500

2020-01-08 31.580000 33.232666 31.215334 32.809334 32.809334 467164500

Let's compute MACD:

```
macd_object = ta.trend.MACD(data['Close'])
data['MACD'] = macd_object.macd()
data['MACD_Signal'] = macd_object.macd_signal()
data['MACD_Diff'] = macd_object.macd_diff()
data.tail()
```

Date	Open	High	Low	Close	Adj Close	Volume	MACD	MACD_Signal	MACD_Diff
2022-12-23	126.370003	128.619995	121.019997	123.150002	123.150002	166989700	-17.244768	-13.664886	-3.579883
2022-12-27	117.500000	119.669998	108.760002	109.099998	109.099998	208643400	-19.072995	-14.746507	-4.326487
2022-12-28	110.349998	116.269997	108.239998	112.709999	112.709999	221070500	-20.000032	-15.797212	-4.202820
2022-12-29	120.389999	123.570000	117.500000	121.820000	121.820000	221923300	-19.771700	-16.592110	-3.179590
2022-12-30	119.949997	124.480003	119.750000	123.180000	123.180000	157777300	-19.258998	-17.125488	-2.133511

```
class ta.trend.MACD(
    close: pandas.core.series.Series,
    window_slow: int = 26, window_fast: int = 12,
    window_sign: int = 9, fillna: bool = False
)
```

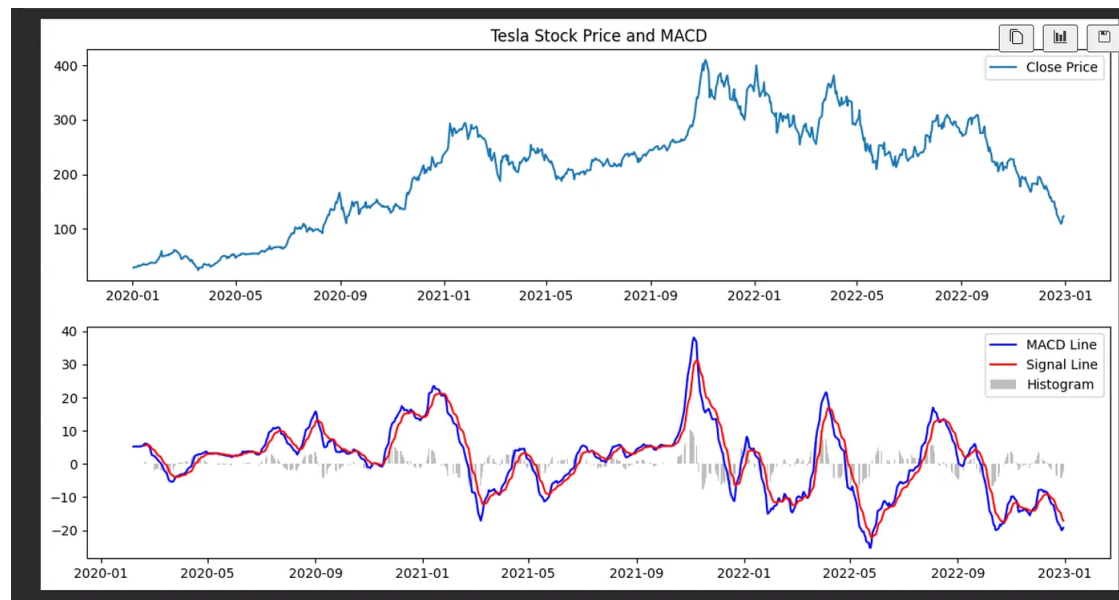
- **close** (*pandas.Series*) — dataset 'Close' column.
- **window_fast** (*int*) — n period short-term.
- **window_slow** (*int*) — n period long-term.
- **window_sign** (*int*) — n period to signal.
- **fillna** (*bool*) — if True, fill nan values.

```
# Plotting
plt.figure(figsize=(14, 7))

# Plotting Close Price
plt.subplot(2, 1, 1)
plt.plot(data['Close'], label='Close Price')
plt.title('Tesla Stock Price and MACD')
plt.legend()

# Plotting MACD
plt.subplot(2, 1, 2)
plt.plot(data['MACD'], label='MACD Line', color='blue')
plt.plot(data['MACD_Signal'], label='Signal Line', color='red')
plt.bar(data.index, data['MACD_Diff'], label='Histogram', color='grey', alpha=0.5)
plt.legend()

plt.show()
```



Plot. Image by the author.

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Python Bollinger Tutorial

towardsdev.com

Python Trading Guide: Leveraging RSI

Python RSI Tutorial

python.plainenglish.io

Time Series Analysis: Mastering the Concepts of Stationarity

Unpacking the Mysteries of Stationarity

python.plainenglish.io

Spearman's Rank Correlation

Simplified Spearman's Rank Correlation in Python

towardsdev.com

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Sources

Documentation - Technical Analysis Library in Python 0.1.4 documentation

Moving average designed to account for market noise or volatility. KAMA will closely follow prices when the price...

technical-analysis-library-in-python.readthedocs.io

MACD (Moving Average Convergence/Divergence Oscillator)

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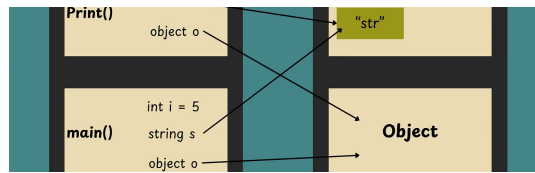


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